Philosophy, Logic and Human Existence

Introduction to Philosophy

Definition, Nature, and Scope of Philosophy

Philosophy is derived from the Greek words *philo* (love) and *sophia* (wisdom), meaning "love of wisdom." It is a discipline that involves the systematic study of fundamental questions about existence, knowledge, values, reason, mind, and language. Philosophy is both a theoretical and practical field, engaging in deep reflection on life's most profound questions.

Nature of Philosophy: Philosophy is characterized by its critical, systematic, and reflective nature. It is concerned with asking and answering deep, fundamental questions about the world and human existence. The nature of philosophy includes:

- **Critical Thinking:** Philosophy challenges established beliefs and promotes logical reasoning and inquiry.
- **Systematic Inquiry:** It employs structured methods to explore fundamental issues.
- **Universality:** Philosophy deals with concepts that are relevant across different cultures and times.
- **Reflectiveness:** Encourages self-examination and the questioning of assumptions.

Scope of Philosophy: Philosophy is a broad discipline that influences various aspects of human life, including science, politics, ethics, religion, and art. Its scope covers multiple areas of human inquiry, and it provides frameworks for analyzing complex problems and developing reasoned solutions.

Branches of Philosophy

Philosophy is generally divided into five major branches, each dealing with different aspects of human thought and existence:

1. **Metaphysics** – The study of reality, existence, and the fundamental nature of being. It explores questions such as: What is real? What is the nature of existence? What is the relationship between mind and matter?
2. **Epistemology** – The study of knowledge, its nature, sources, and limits. It investigates questions like: What can we know? How do we acquire knowledge? What distinguishes justified belief from opinion?
3. **Ethics** – The study of moral principles that govern human behavior. It examines concepts of right and wrong, justice, virtue, and moral responsibility.
4. **Logic** – The study of reasoning, argumentation, and the principles of valid inference. Logic helps distinguish between valid and fallacious reasoning.
5. **Aesthetics** – The study of beauty, art, and taste. It seeks to understand what makes things aesthetically pleasing and how artistic expression impacts human experience.

Importance of Philosophy in Human Life

Philosophy plays a crucial role in shaping human thought, decision-making, and problem-solving.

1. **Enhancing Critical Thinking** – Philosophy trains individuals to analyze arguments, identify biases, and think logically. This skill is essential in making well-reasoned decisions in both personal and professional life.
2. **Moral and Ethical Guidance** – Ethical philosophy helps individuals and societies establish moral values and frameworks that govern behavior.
3. **Understanding Human Existence** – Through existential and metaphysical inquiry, philosophy helps people seek meaning, purpose, and a better understanding of their place in the world.
4. **Improving Communication Skills** – Engaging in philosophical discourse enhances clarity in reasoning, persuasion, and articulation of ideas.
5. **Influencing Other Disciplines** – Philosophy underpins many other fields, including law, politics, science, and technology, by providing foundational principles and methods of inquiry.
6. **Promoting Social and Political Awareness** – Political philosophy informs discussions on justice, rights, governance, and the responsibilities of individuals within a society.
7. **Contributing to Technological and Scientific Advancements** – The philosophy of science examines the methods, assumptions, and ethical implications of scientific discoveries and technological innovations.

Characteristics of a Philosopher

1. **Critical and Rational Inquiry** – Philosophy involves logical reasoning and questioning rather than relying on assumptions or dogma.
2. **Systematic and Structured Thinking** – Philosophical arguments follow structured methods of reasoning and analysis.
3. **Fundamental Questions** – Philosophy addresses deep questions about existence, knowledge, morality, reality, and human purpose.
4. **Universality and Generality** – Unlike specialized sciences, philosophy seeks broad, foundational principles that apply universally.
5. **Conceptual Analysis** – Philosophers analyze the meanings, implications, and interconnections of concepts.
6. **Reflective and Self-Examinatory** – Philosophy encourages self-awareness and critical examination of one's beliefs and assumptions.
7. **Interdisciplinary Nature** – It intersects with other disciplines such as science, religion, politics, and ethics.
8. **Search for Wisdom** – Philosophy is not just about acquiring knowledge but seeking wisdom in understanding life and the universe.
9. **Dialectical Method** – Philosophical discussions often involve argumentation, counterarguments, and refinement of ideas through dialogue.
10. **Normative and Evaluative** – Philosophy not only describes reality but also prescribes values and ethical standards.

2. The Origins and History of Philosophy

- **Early beginnings: African, Eastern, and Western traditions**

Philosophy originated as a product of man's gift of rationality which aids him in his interaction with nature, the environment and human experiences (Molokwu, 2009). Though traces of the origin always start with the western traditions, particularly, the Greeks, its presence in Africa and the Eastern part of the world cannot be overlooked. The early philosophical traditions of Africa, the East (including China and India), and the West (particularly Greece) reveal diverse philosophical thoughts that shaped human intellectual history.

In **Africa**, Egyptian civilization which is one of the oldest in human history, played a role in the development of early philosophical thought as Egyptian philosophy which was mixed with mythology, ethics, and governance, emphasised the concept of Ma'at—order, justice, and balance (Ebo, 2018). Philosophers such as Ptahhotep, who authored "The Maxims of Ptahhotep," provided ethical guidance on leadership, humility, and social harmony. Moreover, recent scholarship suggests that early Greek philosophers, including Thales and Pythagoras, were influenced by Egyptian thought as the Greeks borrowed heavily from Egyptian metaphysics and mathematical principles (Anakwue, 2018). Beyond Egypt, African philosophy thrived in oral traditions and the wisdom of sages. The Akan, Yoruba, and Igbo peoples had sophisticated systems of thought, often transmitted through proverbs, folklore, and storytelling (Kanu, 2013).

In **Eastern** part of the world, the Indian philosophical tradition dates back to the Vedas (c. 1500 BCE), which contain hymns and metaphysical speculations about the nature of existence and the universe. The Upanishads (800–200 BCE) further developed concepts such as Brahman (ultimate reality) and Atman (self), forming the basis of Vedantic thought (Osborne, 1992). According to Osborne (1992), six major philosophical schools—Nyaya, Vaisheshika, Samkhya, Yoga, Mimamsa, and Vedanta—offered different approaches to epistemology, logic, and ethics. Nonetheless, the emergence of Buddhism and Jainism in the 6th century BCE introduced new perspectives on suffering, karma, and liberation. Furthermore, in China, early philosophical thought was shaped by Confucianism, Daoism, and Legalism. Confucius (551–479 BCE) focused on moral virtue, social harmony, and the role of education in personal development. His Analects remain central to Chinese thought (Osborne, 1992). Daoism, attributed to Laozi, offered a contrasting view that valued natural spontaneity and harmony with the Dao (Way). Meanwhile, Legalism, championed by Han Feizi, focused on strict laws and governance as essential for societal order.

Western philosophy is said to have started with wonder and curiosity about the nature of the universe (Molokwu, 2009). It is often traced to the Pre-Socratic thinkers of Ionia, such as Thales, Anaximander, and Heraclitus, who sought natural explanations for the cosmos, moving away from mythological narratives (Ebo, 2018). The achievements of these philosophers will be discussed later.

- **Key philosophical movements: Ancient, Medieval, Modern, and Contemporary**

Philosophy, as a discipline, has undergone transformations over centuries and has been shaped by cultural, political, and scientific developments. Therefore, this sub-section looks into four key philosophical movements: Ancient, Medieval, Modern, and Contemporary philosophy, as each period contributed unique perspectives and helped in shaping the evolution of thought and influencing modern philosophical inquiries.

I. Ancient Period

Ancient philosophy which emerged in the 6th century BCE, focused on understanding the nature of reality, ethics, and knowledge. According to Molokwu (2009), this period can be divided into two: The pre-Socratic and Socratic periods.

The pre-Socratic period focused more on issues related to cosmology—the universe and its constitutive elements and the philosophers of the period were known as members of the Milesian school (Molokwu, 2009). Some of the philosophers of this period include philosophers such as Thales, Anaximander, and Heraclitus, who initiated inquiries into metaphysics and cosmology (Sokal, Oden, Rosenberg, & Digiovanni, 1997).

Thales, the first member of this school, contended that the source or origin of the universe was water as he believed that all things are made from it. Thales' contributions spanned various fields, including philosophy, science, mathematics, and astronomy. Despite the scarcity of primary sources, much of what is known about Thales comes from later philosophers, particularly Aristotle, who credited him as the first philosopher to propose a naturalistic explanation for the cosmos (Cantor, 2022).

Thales' philosophical approach was a radical departure from traditional mythological narratives. He proposed that water was the underlying principle (arche) of all matter, suggesting that the world's complexity could be reduced to a single, fundamental substance. This concept of material monism introduced a new method of inquiry based on natural causes rather than supernatural explanations. Kartal (2023) discusses how Aristotle interpreted Thales' theory as an early attempt at material monism, a concept that would later influence the development of metaphysics. While Aristotle critiqued Thales' idea for its simplicity, he acknowledged its importance in initiating the tradition of seeking naturalistic explanations for natural phenomena.

One of Thales' most significant contributions was his pioneering approach to understanding natural events through observation and reason. He is credited with predicting a solar eclipse which demonstrates his application of empirical methods to astronomy (Molinari, 2022). This achievement not only showcased his knowledge of celestial patterns but also illustrated his departure from viewing such phenomena as governed by capricious gods. According to Rajiyung (2024), Thales' work marked the beginning of the shift from mythos (myth) to logos (reason), a critical transformation that laid the groundwork for scientific inquiry and the philosophical tradition of rational analysis.

Thales' influence extended beyond natural philosophy into mathematics. He is attributed with several geometric principles, including what is now known as Thales' Theorem, which states that if a triangle is inscribed in a circle with one side as the diameter, then the triangle is a right

triangle. Kvasz (2020) argues that Thales' approach to geometry reflected a cognitive unity between his philosophical and mathematical thinking. His use of deductive reasoning in mathematics parallels the methodological rigor he applied to natural philosophy.

After Thales, **Anaximander**, an evolutionist, came up as the second member of this school. He stated that all things came as a product of the activity of an element whose nature is infinite, eternal and ageless (Molokwu, 2009). As a member of the Milesian school, alongside Thales and Anaximenes, Anaximander sought natural explanations for the origin and structure of the cosmos, departing from the mythological frameworks that dominated ancient Greek culture. His ideas introduced the concept of the *apeiron* (the "boundless" or "infinite"), a revolutionary notion that not only influenced subsequent philosophers but also laid foundational principles for scientific inquiry and metaphysical speculation (Stella & Divino, 2023).

Anaximander's concept of the *apeiron* is one of his most enduring philosophical contributions. Unlike his predecessor Thales, who posited water as the primary substance of all things, Anaximander proposed that the cosmos originated from the *apeiron*, an indefinite and boundless principle that transcends conventional elements like water, air, or fire (López, 2020). The *apeiron* was not a specific material substance but rather an eternal and ageless source from which all things emerge and into which they return. This cyclical process of generation and destruction reflects a form of natural justice (*dike*), where the elements atone for their excesses over time, maintaining balance in the universe (Griffith, 2022). By introducing the *apeiron*, Anaximander advanced a metaphysical idea that suggested the world is governed by abstract principles rather than capricious deities, which is a shift towards rationalism and naturalism in Greek philosophy.

Anaximander also made substantial contributions to cosmology. He was the first known philosopher to conceptualise the universe with a structured, mechanical order. According to Naddaf (2019), Anaximander proposed that the Earth was a cylindrical shape, suspended freely at the center of the cosmos, not supported by any physical structure. This idea broke away from earlier mythological views that imagined the Earth resting on a physical foundation, such as the back of a giant creature or a world-pillar. Anaximander's assertion that the Earth remained in its place due to its equidistant position from all points of the celestial sphere introduced a rudimentary concept of equilibrium.

Furthermore, Anaximander's work in astronomy and meteorology revealed his empirical approach to natural phenomena. He is credited with creating one of the earliest maps of the known world and contributing to the understanding of celestial bodies' movements (Zhu, 2024). His theories on the formation of the Earth and the heavens were not only speculative but also grounded in observable phenomena, as seen in his explanation of natural events like rain, wind, and the changing seasons. Rahman (2023) highlights Anaximander's use of dialectical reasoning, particularly his exploration of opposites—such as hot and cold, wet and dry—as fundamental forces driving natural change. This dialectical method influenced later philosophical traditions, including Heraclitus' doctrine of change and the development of dialectics by Plato and Aristotle.

Anaximander's philosophical ideas also extended into the realm of biology and the natural sciences. He proposed an early theory of evolution, suggesting that life originated in the moisture of the Earth and that human beings evolved from fish-like creatures (Salles, 2024). This proto-evolutionary theory demonstrated his ability to apply naturalistic explanations to biological processes. His ideas on natural processes emphasised the importance of observation and rational analysis.

Anaximenes, the third of the Milesian school, differed in opined from both Thales and Anaximander as he was concerned with the nature of cosmic matter or primary element (Ozuzu, 2005). Anaximenes proposed that air was the fundamental substance from which all things originate. Unlike his predecessor Anaximander, who posited the *apeiron* (the "boundless") as the ultimate principle, Anaximenes chose a more tangible and observable element. He introduced the theory of condensation and rarefaction (Molokwu, 2009). He argued that air could transform into various forms of matter through processes of condensation and rarefaction. According to Hobza (2020), Anaximenes suggested that when air condenses, it becomes visible and material, transforming into water, then earth, and finally stones. Conversely, when air rarefies, it becomes fire. This theory provided an early model of physical transformation based on natural processes, which shows Anaximenes' pioneering approach to explaining the physical world through empirical observation and rational deduction.

Anaximenes' understanding of air extended beyond its material aspect to a more metaphysical interpretation. He viewed air as a life-giving force, drawing an analogy between the macrocosm of the universe and the microcosm of human life. As air sustains human life through breathing, so too does it sustain the universe. Deng (2023) points out that this perspective infused Anaximenes' philosophy with a quasi-spiritual dimension, as air was not only a physical substance but also a source of vitality and motion.

Anaximenes' contributions to cosmology were also significant as he envisioned a flat Earth suspended in air, with celestial bodies such as the sun, moon, and stars floating above it (Kočandrle, 2019). He proposed that these heavenly bodies were ignited by the movement of air and that their rising and setting were optical illusions caused by their rotation around the Earth. This meteorological approach to cosmology differed from modern scientific understandings but represented a critical step toward naturalistic explanations of celestial phenomena. By attributing cosmic events to observable natural processes, Anaximenes laid the groundwork for later astronomical theories and challenged the notion that such events were governed by the whims of gods.

Deng (2023) draws a comparison between Anaximenes' concept of air as a fundamental principle and the Chinese philosophy of *Qi* as described in the *Wuxing* (Five Elements) framework. Both systems propose a dynamic, process-oriented understanding of the natural world, where a basic substance transforms into different forms through specific mechanisms. Such cross-cultural analyses underscore the universality of early human attempts to understand the natural world through elemental theories, suggesting that Anaximenes' thought was part of a broader pattern of philosophical inquiry into the nature of existence.

Anaximenes' methodology also reflects an evolution in scientific thinking. His emphasis on natural processes like condensation and rarefaction can be seen as an early form of empirical science, where hypotheses about natural phenomena were grounded in observable processes (Hobza, 2020). While Anaximenes lacked the experimental rigor of modern science, his approach contributed to the development of the scientific method by promoting the idea that natural laws governed the behavior of matter. His influence extended to later philosophers, including Heraclitus, who developed the concept of perpetual change, and Empedocles, who proposed a more developed theory of the four classical elements.

Pythagoras was also an Ionia but migrated to kroton in southern Italy. He formed a movement known as the pythagorean school (Ozuzu, 2005). Pythagoras was a genius at calculation, the shapes of symbols and he calculated the triangle. He also believed in reincarnation and claimed that the soul was divine and undergoes purification through life, death, and rebirth to attain perfection (Molokwu, 2009). Despite the scarcity of direct writings attributed to Pythagoras, his legacy is preserved through the works of later philosophers such as Plato, Aristotle, and Iamblichus (Moore, 2019).

Central to Pythagorean philosophy is the concept that numbers constitute the fundamental essence of reality. Pythagoras and his followers believed that "all things are numbers," suggesting that the universe's structure and harmony could be understood through mathematical relationships (Amangeldiyev & Aldanazarova, 2020). This perspective laid the groundwork for the metaphysics of numbers, where numbers were not merely tools for counting but intrinsic elements of the cosmos with mystical and ontological significance. The Pythagoreans proposed that the principles of mathematics could explain natural phenomena, with particular emphasis on the ratios found in musical harmony. Konovalova (2021) reveals how Pythagoras' discovery of the harmonic intervals in music—such as the octave, fifth, and fourth—demonstrated the deep connection between numerical ratios and the physical world, leading to the concept of the "harmony of the spheres." This idea proposed that celestial bodies produced a form of cosmic music through their movements, reflecting a universal order governed by mathematical laws.

Pythagoras' contributions to mathematics extend beyond metaphysics. He is best known for the Pythagorean theorem, which establishes the relationship between the sides of a right-angled triangle. While this theorem was likely known to earlier cultures, Pythagoras is credited with providing the first formal proof, a hallmark of Greek mathematical rigor (Erovenko & Michailova, 2022). His approach to mathematics emphasised logical deduction and proof, influencing the development of geometry and laying the foundations for Euclidean geometry. Chiotis (2020) opines that Pythagorean mathematical principles influenced ancient architecture, including the design of the Polycratean temple of Heraion in Samos. The use of Pythagorean triples in temple layouts also illustrates how Pythagorean mathematics was not only theoretical but also applied in practical contexts.

The Pythagorean school also ventured into metaphysical and religious domains. Pythagoras introduced the doctrine of metempsychosis, or the transmigration of souls, which posits that the soul is immortal and undergoes a cycle of rebirths (John & Akpan, 2021). This belief, influenced by Orphic traditions, emphasises the ethical dimensions of Pythagoreanism, which advocates for

a life of purity and self-discipline to achieve spiritual liberation. De Campos (2022) opines that Pythagoras' unwritten teachings, preserved through oral tradition and interpreted by later Neoplatonists like Hermias of Alexandria, contributed to a mystical strand of Greek philosophy that valued esoteric knowledge and the inner transformation of the soul. This integration of philosophy with religious practice positioned Pythagoras as not merely a mathematician or scientist but as a spiritual guide whose teachings aimed at the cultivation of both mind and spirit.

Pythagorean philosophy also extended into the realm of political thought. According to Brovkin (2024), the early Pythagoreans promoted political stability, moderation, and the integration of philosophical principles into public life. Their community in Croton, Southern Italy, functioned as a model of philosophical governance, emphasising equality, justice, and communal living. However, Pythagorean influence in politics was not without controversy. The secretive nature of their society and their elitist inclinations led to political conflicts, resulting in the persecution of Pythagoreans and the dispersal of their community. Despite this setback, the political ideals of Pythagoras influenced later thinkers, including Plato, whose Republic echoes Pythagorean notions of philosopher-kings and the role of education in achieving a just society.

Pythagoras' influence is also evident in his impact on music theory and cultural works. Konovalova (2021) explains that Pythagoras' insights into the mathematical basis of musical harmony contributed to the mathematisation of music, transforming it from a mystical art into a disciplined science. This approach laid the groundwork for Western music theory and reinforced the idea that beauty and harmony in art are reflections of deeper mathematical and cosmic orders. Moreover, Danesi (2020) emphasises Pythagoras' enduring legacy in the intellectual traditions of science and philosophy, arguing that his integration of mathematics, music, and metaphysics established a model of interdisciplinary inquiry that continues to inspire contemporary thinkers.

Another influential philosopher of this period was **Heraclitus**, who argued that nothing is permanent in this world except for change (Molokwu, 2009). Heraclitus also added that in spite the fact that there is perpetual change in the world, there is also perpetual conflict and strife (Ozuzu, 2005). Heraclitus of Ephesus (c. 535–475 BCE) was known for his doctrine of flux and his enigmatic aphorisms. His philosophy, characterised by the idea of constant change and the unity of opposites, contributed significantly to the development of Western philosophy, particularly through his concept of the *logos* (reason or word). Heraclitus' work, primarily surviving in fragments, presents a worldview where stability is an illusion, and change is the only constant (Desta, 2023).

Heraclitus' doctrine of flux is famously encapsulated in his statement that one cannot step into the same river twice, which reveals his belief that everything is in a state of continuous transformation. According to Hui (2019), this idea contrasts sharply with later philosophical systems, such as Plato's theory of eternal Forms, which emphasise stability and permanence. Heraclitus argued that all things are in a perpetual state of becoming, driven by an underlying order he called *logos*. This *logos* is not only a principle of order but also a rational structure that governs the cosmos, suggesting that while things change, they do so according to a rational pattern (Hui, 2019).

The concept of *logos* plays a central role in Heraclitus' philosophy. He proposed that the *logos* is an objective truth accessible through reason, distinguishing it from the mythical explanations of the world offered by poets like Homer and Hesiod (Desta, 2023). Halapsis (2020) emphasises that Heraclitus' *logos* bridges the gap between mythos and logos as it represents a shift from mystical explanations to rational thought.

Heraclitus is also known for his principle of the unity of opposites. He believed that opposites are not mutually exclusive but interdependent. For example, he argued that life and death, day and night, and war and peace are interconnected, and one cannot exist without the other (De Campos, 2022). This principle led to his famous assertion that "war is the father of all things," suggesting that conflict and tension drive the creation and destruction that sustain the cosmos (Lopes, 2021). Heraclitus' exploration of opposites laid the groundwork for dialectical thinking, later developed by philosophers such as Hegel and Nietzsche.

Heraclitus' philosophy of fire as the fundamental element of the cosmos is another key aspect of his thought. He viewed fire not only as a physical substance but as a metaphor for change and transformation. Ignatenko (2020) explains that for Heraclitus, fire represented the ever-present process of creation and destruction, mirroring the perpetual flux of the universe.

Heraclitus' enigmatic style and use of paradox have also contributed to his enduring legacy. His writings are often described as obscure, using metaphor, symbolism, and aphoristic statements that resist straightforward interpretation (Jeck, 2019).

Parmenides from Elea, was the first to focus on the difference between the senses and reason (Molokwu, 2009). According to Ozuzu (2005), the philosophy of Parmenides is contained in his poem title 'On Nature,' in which he explained two things, the way of truth and way of opinion. According to Parmenides, the first is the way of reason while the second is the of the way of the senses (Ozuzu, 2005). Central to Parmenides' philosophy is the assertion that "what is, is" and "what is not, is not," encapsulating his principle of the immutability and unity of being. He argued that being is ungenerated, indestructible, indivisible, and unchanging. According to Meza (2020), Parmenides' monism posits that reality is a singular, continuous entity, and any perception of change or plurality is illusory. This view directly opposed earlier thinkers like Heraclitus, who emphasised the constant flux of the universe. Parmenides' rejection of change and multiplicity led to a stark dichotomy between the world of appearances and the world of true reality, influencing later philosophers such as Plato and Aristotle.

The method by which Parmenides arrived at his conclusions involves rigorous logical reasoning, which some scholars consider an early form of deductive logic. His poem introduces the "Way of Truth," where he outlines the characteristics of being, and contrasts it with the "Way of Opinion," which deals with the deceptive nature of sensory experience. Sorensen (2022) highlights that Parmenides' approach to truth is not empirical but rational, relying on the intellect to access an eternal and unchanging reality. This rationalist perspective marked a critical shift from mythological explanations of the cosmos to a more systematic and philosophical approach to knowledge.

Parmenides' influence on the development of logic is also noteworthy. Çetin (2022) argues that Parmenides' work prefigures several principles of classical logic, including the law of non-contradiction, the principle of identity, and the law of the excluded middle. These principles, though not formally articulated until Aristotle, are implicit in Parmenides' arguments against the possibility of non-being and change. His use of logical consistency to support his ontological claims set a precedent for the philosophical tradition of constructing metaphysical theories through deductive reasoning.

Despite Parmenides' emphasis on the unity of being, his poem also addresses cosmological ideas by creating an apparent contradiction between his metaphysical monism and his descriptions of the physical world. Conte (2023) discusses this paradox, noting that while Parmenides denies the reality of becoming and change in the "Way of Truth," he provides a cosmological account in the "Way of Opinion" that includes elements of traditional Greek cosmology, such as the roles of light and night. Scholars have debated whether this cosmological section represents a concession to popular belief or a didactic tool to illustrate the errors of sensory perception. This dual strategy, as suggested by Thanassas (2020), might reflect a deeper methodological approach, where Parmenides juxtaposes truth with illusion to reinforce his argument about the unreliability of the senses.

There were also the Atomists and Sophists. **Atomists** like Leucipus and Democritus believed that everything is composed of atoms which are physically indivisible and indestructible while **Sophists** were teachers who employed the deductive method often beginning with particular cases and ending with general principle (Molokwu, 2009).

The Socratic period is the second period of the ancient period in the historical development of philosophy. Socrates did not write anything (Molokwu, 2009); however, Bonacchi, Witte, & Altaweel (2024) reveal that Socrates introduced ethical philosophy which emphasised rational inquiry and dialectics and his student, Plato, expanded on these ideas, advocating for idealism and establishing the Academy. According to Molokwu (2009), Socrates compelled philosophy to consider lifestyle, issues of existence and morals. Thus, one of Socrates' most significant contributions to philosophy is his development of the Socratic method, a form of dialectical questioning designed to stimulate critical thinking and expose contradictions in thought. This method involves asking a series of probing questions to help individuals examine their beliefs and arrive at a deeper understanding of truth. According to Avdić (2021), this approach marked a critical transition from pre-Socratic natural philosophy, which focused on explaining the physical world, to a new focus on human life, ethics, and the pursuit of virtue. Socrates believed that through dialogue and questioning, individuals could achieve self-knowledge, which he considered the highest form of wisdom. His famous assertion, "I know that I do not know," highlights his commitment to intellectual humility and the endless pursuit of knowledge (Avdić, 2021).

Socrates' philosophy was not only methodological but also ethical. He is often credited as the first philosopher to place ethics at the center of philosophical inquiry. Wolfsdorf (2020) argues that Socrates' ethical philosophy was revolutionary in its emph asis on the idea that virtue is knowledge. According to this view, wrongdoing is a result of ignorance, and those who truly

understand what is good will naturally do good. This perspective introduced a rational basis for ethics, suggesting that moral excellence could be taught and learned through education and reflection. Socrates' ethical views also laid the groundwork for later philosophical traditions, including stoicism and Christian ethics, which similarly emphasise the importance of wisdom and the cultivation of the soul.

The life and death of Socrates serve as a powerful testament to his philosophical principles. In 399 BCE, he was tried and sentenced to death by the Athenian state on charges of impiety and corrupting the youth. His trial and defense, as recorded in Plato's Apology, show his unwavering commitment to his philosophical mission. Jiao (2023) notes that Socrates used his trial as an opportunity to educate the public about the importance of virtue and the examined life. Rather than plead for his life, he maintained that a true philosopher must not fear death but instead remain dedicated to the pursuit of truth and justice. His acceptance of the death penalty rather than compromising his beliefs solidified his status as a martyr for philosophy, embodying the ideal of living according to one's principles regardless of the consequences.

Socrates' impact on subsequent philosophy cannot be overstated. Taylor (2019) emphasises that Socrates' influence on Plato was particularly significant as Plato's portrayal of Socrates not only preserved his teachings but also transformed them into a robust philosophical framework that underpinned the development of Western philosophy. Through Plato's dialogues, Socratic ideas influenced a wide array of philosophical schools, including the Academy founded by Plato, Aristotelian philosophy, and later, the Stoics, who adopted Socratic ethics as a core component of their doctrine. Socrates' ideas also reached into medieval thought through the revival of Platonic philosophy during the Renaissance.

Plato (c. 427–347 BCE), a student who was greatly influence by his teacher, Socrates, contributed immensely to the development of philosophy. To him, man is made of the soul (reason) and the body (Molokwu, 2009). Plato introduced the concept of Forms (or Ideas), proposing that the material world is a mere shadow of a higher, unchanging reality of abstract, perfect entities. According to Nicolae Sfetcu (2022), Plato distinguished between perceptible but unintelligible reality (empirical knowledge) and imperceptible but intelligible reality (rational knowledge) by drawing influence from Pythagoras and mathematics. Plato emphasised the role of reason in acquiring knowledge. He argued that true knowledge (episteme) comes from rational insight rather than sensory experience (doxa or mere opinion). His dialectical method, based on structured argumentation, remains central to philosophical inquiry (Herrmann, 2007). Plato's ethical theory was rooted in his concept of the tripartite soul, consisting of reason, spirit, and appetite. He also argued that Justice emerges when these elements function harmoniously. Plato also made contributions to mathematical philosophy and its application in natural sciences. Wu Wen (2013) explains that he integrated mathematical reasoning into his metaphysical and scientific outlook, influencing fields such as astronomy. Plato founded the Academy in Athens, one of the first higher-learning institutions in the Western world (Ahemad & Shah, 2016).

Aristotle, Plato's student, advanced empiricism by laying the groundwork for scientific methodology. His works in ethics, politics, and metaphysics influenced various disciplines (Narasimhan et al., 2019). Molokwu (2009) reveals that he founded a school called Lyceum and

was known to have developed formal logic which deals with the form that reasoning takes. One of Aristotle's significant contributions to philosophy is his work in metaphysics, where he explored the nature of being and reality. His seminal text, *Metaphysics*, introduces the concept of 'substance' as the primary category of being. Aristotle argued that substance is composed of both matter and form, where matter is the potentiality and form is the actuality (Meister, 2022). He proposed that the ultimate reality is not separate from the physical world, as Plato suggested, but is present within tangible objects through their forms. Aristotle's doctrine of the "four causes" (material, formal, efficient, and final) provided a comprehensive framework for explaining change and existence in the natural world. His notion of the "unmoved mover" as the first cause of all motion influenced theological and philosophical discourse for centuries, particularly in medieval scholasticism (Jeffrey, 2021).

Aristotle's contributions to logic are foundational to Western philosophy. His *Organon* introduced formal logic and syllogistic reasoning, establishing a method of deductive reasoning that dominated philosophical thought until the modern era (Castelli, 2020). Aristotle's categorisation of logical propositions and his development of the principles of non-contradiction and the excluded middle became critical tools for evaluating philosophical arguments and scientific hypotheses. Aristotle categorised propositions into four types: universal affirmative, universal negative, particular affirmative, and particular negative. Taysina (2023) emphasises that Aristotle's methodological arsenal, particularly his use of categories, provided a structured approach to philosophical inquiry that influenced not only ancient thought but also modern philosophy through figures like Immanuel Kant.

In ethics, Aristotle is best known for his concept of *eudaimonia*, often translated as "happiness" or "flourishing." His 'Nicomachean Ethics' argues that the highest human good is living a life of rational activity in accordance with virtue (Melnikov, 2023). Aristotle distinguished between intellectual virtues, which involve reasoning, and moral virtues, which arise from habit. He introduced the "Golden Mean," a doctrine that virtue lies between the extremes of excess and deficiency. For example, courage is a mean between recklessness and cowardice. Aristotle's ethical theory is a form of virtue ethics, which contrasts with consequentialist and deontological ethical frameworks developed later in Western philosophy. His ideas on ethics were not only theoretical but also practical, influencing political philosophy and education.

Aristotle's political philosophy is articulated in Politics, where he explores the role of the *polis* (city-state) in achieving the good life. He believed that humans are "political animals" whose fulfillment is closely tied to their participation in civic life (Ishino, 2022). Aristotle analysed different forms of government, including monarchy, aristocracy, and democracy and proposed a mixed regime as the most stable and just form of governance. He emphasised the importance of the middle class in maintaining political stability and critiqued both extreme democracy and oligarchy for their potential to lead to tyranny. Unlike Plato, who envisioned a philosopher-king ruling the ideal state, Aristotle advocated for a more pragmatic approach that considered historical and cultural contexts (Trang & Quỳnh, 2021).

Medieval Philosophy

Medieval philosophy, spanning roughly from the 5th to the 15th century, helped in preserving and transforming classical philosophical traditions. This period was influenced by Greek, Roman, Christian, Jewish, and Islamic thought and sought to reconcile faith with reason, laying the foundation for modern philosophy. Following the fall of the Western Roman Empire, philosophical activity shifted towards preserving ancient knowledge and integrating it with Christian theology. Thus, medieval philosophy bridged classical philosophy with theological doctrines. Augustine's works integrated Platonic thought into Christianity (Cesaretti, et al., 2016). One notable philosopher of this period was **St. Augustine (354–430)**. Influenced by Neoplatonism, he laid emphasises on the role of divine illumination in knowledge (Marenbon, 2006).

Boethius (477–524) translated and commented on Aristotle's logic, bridging classical and medieval thought (György, 2013). The monastic tradition in Europe helped preserve and transmit classical works.

The Islamic and Jewish influence (9th–12th Century) was key to the development of philosophy as Islamic and Jewish scholars helped in transmitting and interpreting classical philosophy. Scholars like **Avicenna (Ibn Sina) (980–1037)** synthesised Aristotelianism and Neoplatonism, contributing to metaphysics and logic (Inglis, 2002).

Averroes (Ibn Rushd) (1126–1198) emphasised on reason and provided detailed commentaries on Aristotle which influenced Latin Scholasticism (König-Pralong, 2009).

Maimonides (1138–1204) integrated Jewish theology with Aristotelian philosophy which impacted Christian thought.

Furthermore, in 12th–14th Century, there was the translation movement in Islamic Spain which reintroduced Aristotle's works to Christian Europe, leading to Scholasticism. This era of scholasticism had scholars like **St. Anselm (1033–1109)** who developed the ontological argument for God's existence (Deely, 2010).

Thomas Aquinas (1225–1274) merged Aristotelian philosophy with Christian theology, emphasising natural reason and divine revelation (Schmiga & Porro, 2013). He synthesised advocated for natural law and rational theology. His work remains a cornerstone of Catholic philosophy (Musielak, Kubicka, Woźniak, Jóźwiak, & Liu, 2021). **John Duns Scotus (1266–1308)** and **William of Ockham (1287–1347)** challenged Aquinas, introducing Nominalism, which questioned the reality of universals (Pomplun, 2016).

In the late Medieval period (14th–15th Century) William of Ockham proposed the Ockham's Razor, which emphasised simplicity in philosophical reasoning, anticipating scientific empiricism (Risti & Gorgiev, 2006). Also, the decline of Scholasticism and the rise of Renaissance humanism shifted the focus from theological concerns to human-centered inquiry. The revival of Platonic and Aristotelian traditions in the Renaissance set the stage for modern philosophy (Gilson, 2019).

II. Modern Philosophy

Modern philosophy (17th-19th century) shifted towards individual reason and empirical observation. It marked a transition from medieval scholasticism to new ways of understanding reality, knowledge, and human existence. During this period, the Age of Reason, scientific advancements, and political revolutions greatly influenced philosophical thought, leading to Rationalism, Empiricism, Idealism, Existentialism, and Analytic Philosophy. Descartes' cogito argument laid the foundation for rationalism, while Locke and Hume developed empiricism (Bašić, Kružić, Jerković, Anđelinović, & Anđelinović, 2017).

A. The Early Modern Period (17th–18th Century)

In the early modern period, there was focus on Rationalism which emphasised reason as the source of knowledge. Some scholars of rationalism this period includes:

- **René Descartes (1596–1650)** introduced methodological skepticism and the famous dictum "Cogito, ergo sum" (I think, therefore I am) (Shapiro, 2016).
- **Baruch Spinoza (1632–1677)** and **Gottfried Leibniz (1646–1716)** developed monist and pluralist metaphysical systems.

B. Empiricism in this period emphasised experience and sensory perception as sources of knowledge.

- **John Locke (1632–1704)** argued for the mind as a tabula rasa (blank slate).
- **George Berkeley (1685–1753)** and **David Hume (1711–1776)** critiqued materialism and introduced skepticism (Mercer & O'Neill, 2005).
- **Immanuel Kant (1724–1804)** synthesised rationalism and empiricism in his "Critique of Pure Reason," arguing that knowledge is shaped by both sensory experience and innate structures of reason (Corrias, 2019). He introduced the concept of synthetic a priori knowledge (Eraslan, 2014).

C. German Idealism and the 19th Century

German idealism scholars in the 19th century includes:

- **Georg Wilhelm Friedrich Hegel (1770–1831)**, who developed dialectical idealism, where reality unfolds through contradictions and resolutions (Bunge, 2014).
- **Arthur Schopenhauer (1788–1860)** introduced pessimism, seeing life as driven by irrational "Will".
- **Karl Marx (1818–1883)** applied dialectics to history, proposing materialism and class struggle as central forces shaping society [(Liao, 2000).

D. Existentialism and Phenomenology (19th–20th Century)

- **Søren Kierkegaard (1813–1855)** and **Friedrich Nietzsche (1844–1900)** rejected absolute truths and focus on individual existence, authenticity, and the "death of God".
- **Edmund Husserl (1859–1938)** developed Phenomenology by emphasising conscious experience as the foundation of knowledge.

- **Jean-Paul Sartre (1905–1980)** and **Martin Heidegger (1889–1976)** expanded Existentialism, arguing for radical human freedom and responsibility (Han, 2006).

E. Analytic Philosophy and 20th-Century Developments

- **Ludwig Wittgenstein (1889–1951)** argued that language shapes reality. He divided his work into early logical positivism and later ordinary language philosophy.
- **Bertrand Russell (1872–1970)** and **G.E. Moore (1873–1958)** critiqued Idealism, laying the groundwork for modern Analytic Philosophy.
- Postmodernism, led by **Michel Foucault (1926–1984) and Jacques Derrida (1930–2004)**, deconstructed truth, power, and language, challenging the foundations of Western philosophy (Sevostyanov, 2023).

III. Contemporary Philosophy

Contemporary philosophy emerged in the late 19th and 20th centuries and has been shaped by scientific advancements, globalisation, and interdisciplinary influences. It includes movements such as Analytic Philosophy, Continental Philosophy, Postmodernism, and Political Philosophy, all of which have engaged with modern social, ethical, and epistemological issues (Bartels, Onstenk, & Veugelers, 2016). This period emphasised historicism, questioning whether philosophy should evolve based on contemporary concerns rather than being tied to historical traditions (Menzhulin, 2022).

A. Analytic Philosophy and Its Influence

Molokwu (2009) revealed that the analytic movement was brought by G. E. Moore and Bertrand Russell. They opined that philosophy should concern itself with analysing language (Molokwu, 2009). The linguistic turn, initiated by Wittgenstein, argued that philosophical problems stem from misunderstandings of language (Frede, 2021). Later pragmatist and post-analytic philosophers, such as W.V.O. Quine and Richard Rorty, challenged foundationalism, emphasizing practical consequences over abstract theorising (Watson, 2012).

B. Continental Philosophy and Postmodernism

Existentialism, with figures like Jean-Paul Sartre and Martin Heidegger, focused on individual freedom and meaning in a post-religious world.

Postmodernism, led by Michel Foucault and Jacques Derrida, deconstructed metaphysical and historical assumptions, emphasising power structures, language, and subjectivity (Jiong, 2006). Furthermore, critical theory, influenced by the Frankfurt School, engaged with Marxist and psychoanalytic ideas to critique ideology and cultural systems.

C. Political and Ethical Philosophy

John Rawls revitalised political philosophy with his theory of justice which emphasised fairness and social contract theory. Feminist and postcolonial philosophy, including bell hooks and Frantz Fanon, critiqued traditional power structures and expanded the discourse on identity, race, and gender (Xing, 2011).

3. Logic: The Science of Correct Thinking

- Definition and nature of logic

Man by nature is a logical being and without logic, man would cease to be man (Ukah, 2010). This provides us with the need to ask questions like what is logic? Of what importance is logic to man? To answer the question of what logic is, it can be noted that it has no universally accepted definition as it has been defined in various ways by different scholars; however, the different definitions point out some salient features of what logic is about (Ogbinaka, 2009). Ogbinaka (2009) reveals that etymologically, logic is derived from the Greek word 'Logos' which means 'word,' 'reason' or 'thought.' Some definitions by scholars are given below:

Irving Copi sees logic as "the study of the methods and principles used to distinguish good (correct) from bad (incorrect) reasonings" (Copi, 1978). Ogbinaka (2009) reveals that Copi further gave two other definitions of logic: "Logic is the science of the law of thought" and "logic is the science of reasoning," which bring it closer to psychology.

A.R. Lacey opines that logic deals with "valid reasoning, its systematisation and the study of notions relevant to it." It has two main parts, formal logic and philosophical logic which is also called logical theory (Lacey, n.d.). Moreover, Mourant defines "logic as both a science and an art, concerned with the quest for knowledge and truth. It is also a study of the validity or correctness of our reasoning" (Nze Agwaranze, 2005).

Leonard describes logic as "the science of exact reasoning." He believes that the science of logic seeks to discover and state laws in accordance with which any act of thought may be judge good or bad (Nze Agwaranze, 2005). Bertrand Russell also defines logic in terms of the formal and legitimate ways of reasoning validly (Russell, 1903).

Ogbinaka (2009) points out the following as important points to be gleaned from various definitions of logic:

(i) Logic concerns orderly reasoning and thought (i.e thinking).
(ii) It is about inferences as it deals on how propositions are inferred.
(iii) It deals with the order and form of arguments.
(iv) It is about judgments and how these judgments are either arrived at or derived.
(v) It is more of a formal, not so much of factual inquiry into how we make our judgments in reasoning.
(vi) Logic can only be the activity of rational human beings since it relates to thought.

The fundamental question of whether logic is a discovery about reality or a human construction has been a long-standing philosophical debate. Logical realism, as advocated by thinkers like A.N. Prior, asserts that logic is not merely a linguistic or conceptual construct but reflects an objective structure of the world (Jakobsen, 2020). This suggests the idea that logical principles govern reality independently of human cognition. Conversely, inferentialist and expressivist views, such as those proposed by Brandom, argue that logic is shaped by human discourse and the rules governing natural and artificial languages (Peregrin, 2019).

Aristotle's contribution remains one of the most enduring aspects of logic, with his syllogisms establishing foundational principles for deductive reasoning (O'Regan, 2020). His work was expanded by the Stoics, who introduced propositional logic, wherein logical statements could be evaluated as true or false independent of their constituent terms. Later developments by thinkers like Frege, Boole, and Russell revolutionised logic, transforming it into a formal system with precise symbolic representation (O'Regan, 2020). Frege's work was also important in shifting logic from a language-dependent system to a formal, axiomatic structure (Shieh, 2019). His conception of logic as a system of truths grounded in primitive axioms provided the basis for modern predicate logic. He argued that logical truths justify themselves through an inherent rational perception, which contrasts with later formalist interpretations that treat logic as an abstract mathematical structure (Shieh, 2019).

In essence, logic, as a discipline, is concerned with the principles of valid inference, the nature of truth, and the structure of rational argumentation. It reveals when, how and if reasoning, judgement and arguments are correct or incorrect; valid or invalid.

- **Types of logic: Formal and Informal logic**

Logic is the foundation of structured reasoning as it provides a systematic method for evaluating arguments and distinguishing valid inferences from fallacious ones. It is basically divided into formal logic and informal logic.

- **Formal Logic**

Formal logic is the study of principles and structures that govern valid reasoning. It focuses on the form of arguments rather than on their content and thus, provides a ground for distinguishing between valid and invalid arguments. It also allows reasoning to be expressed in a precise and systematic manner by using formal symbols and rules.

Historically, formal logic can be traced back to ancient philosophers such as Aristotle, who formulated syllogistic logic to study deductive reasoning. His system was the first attempt to establish a formal structure for logical arguments based on categorical propositions. In the 19th and 20th centuries, logicians such as Frege, Peano, Russell, and Hilbert expanded upon Aristotle's ideas, developing symbolic logic and formal systems that laid the foundation for modern logic (Ono, 2019). The formalisation of logic during this period led to breakthroughs in the foundations of mathematics and the philosophy of logic.

Fundamentally, formal logic operates through well-defined syntactic rules and semantics that determine how statements relate to one another. The syntax of formal logic consists of symbols and rules for combining them into well-formed formulas and these formulas represent propositions, which are declarative statements that can be either true or false. The semantics of formal logic assigns meaning to these formulas by interpreting them within a logical system. This dual structure—syntax and semantics—ensures that formal logic maintains both precision and consistency (Polat, 2024).

Formal logic also relies on axioms and inference rules. Axioms are fundamental logical truths assumed to be self-evident, while inference rules determine how new truths can be derived from

existing ones. Logical systems such as propositional logic and predicate logic are built upon these axioms and rules. Propositional logic, also known as sentential logic, deals with propositions as whole units and focuses on logical connectives such as "and," "or," "if…then," and "not." It evaluates the truth value of compound statements based on the truth values of their components. Predicate logic, or first-order logic, extends propositional logic by incorporating quantifiers ("for all" and "there exists") and predicates, allowing for more complex reasoning about objects and their properties (Polat, 2024).

The precision of formal logic makes it an essential tool in mathematical reasoning because it provides the foundation for mathematical proofs by ensuring that conclusions follow necessarily from premises. Mathematicians use formal logic to construct rigorous arguments, eliminating ambiguity and potential errors in reasoning (Zach, 2024). The formalisation of mathematics has also led to the development of mathematical logic, which deals with the relationship between logic and mathematical structures. However, the application of formal logic extends beyond mathematics and philosophy to incorporate fields such as computer science and artificial intelligence. In computing, formal logic is used to design programming languages, verify algorithms, and develop automated reasoning systems. Logic programming, a paradigm based on formal logic, enables computers to perform logical inference and solve complex problems (Zach, 2024). Additionally, formal logic plays a vital role in artificial intelligence, specifically in knowledge representation and automated theorem proving. As a result, AI systems can reason about data and make decisions based on logical principles through encoding knowledge in a logical format.

- **Informal Logic**

Informal logic is the study of reasoning as it naturally occurs in everyday language, focusing on evaluating arguments in real-world contexts. Unlike formal logic, which relies on symbolic notation and abstract structures to assess validity, informal logic is concerned with how arguments are constructed, interpreted, and evaluated in practical discourse.

Basically, informal logic seeks to understand how people argue outside the rigid constraints of formal systems. It recognises that in daily reasoning, arguments are often influenced by context, emotions, rhetorical techniques, and audience perception (Tindale, 2019). Informal logic instead asks whether an argument is reasonable and convincing given its context, rather than simply whether it follows formal rules of inference. The rise of informal logic as a distinct field was a response to the limitations of formal logic in evaluating everyday reasoning. Philosophers and educators in the mid-20th century began to advocate for a more practical approach to logic that can address reasoning in ordinary language rather than abstract symbols (Kakas, 2019).

One important component of informal logic is the analysis of argument structure. Informal logic often deals with incomplete, implied, or context-dependent reasoning and thus, arguments are evaluated based on their coherence, persuasiveness, and ability to withstand critical scrutiny. This makes informal logic particularly useful in analysing persuasive communication, including political speeches, advertising, legal arguments, and journalistic reports (Polat, 2024). A central concern of informal logic is the identification of logical fallacies—flaws in reasoning that can

make an argument misleading or unsound. Common informal fallacies include ad hominem attacks (attacking the person rather than their argument), straw man arguments (misrepresenting an opponent's position to make it easier to refute), and false dilemmas (presenting only two options when more exist (Gupta et al., 2021).

- **Importance of Logic in Philosophy and Daily Life**

Logic is important in both philosophy and daily life as it provides a structure for reasoning, argumentation, and decision-making. It is the foundation of rational thought which enables individuals to construct sound arguments, analyse complex problems and also distinguish between valid and invalid reasoning.

In philosophy, logic is an essential tool used in evaluating arguments and looking into fundamental questions about reality, knowledge and truth. Philosophers rely on logical principles to assess the validity of arguments and avoid fallacies that may lead to incorrect conclusions. Bertrand Russell described logic as "the essence of philosophy," by highlighting that it clarifies philosophical problems through rigorous reasoning (MacFarlane, 2020). Philosophical logic, particularlly seeks to resolve ambiguities in natural language by analysing logical forms. This method allows philosophers to construct precise arguments and addresses issues in metaphysics, epistemology, ethics and the philosophy of language.

The stoics, including Epictetus, viewed logic as a necessary discipline for achieving wisdom and virtue. They believed that logical reasoning protected the mind from fallacies, enabling individuals to make rational decisions and distinguish between right and wrong [(Durhan, 2022). This perspective emphasises the ethical dimension of logic, suggesting that clear reasoning contributes to moral decision-making and intellectual freedom. Similarly, in modern times, logic continues to play a role in ethical theories, legal reasoning, and debates about moral responsibility.

Beyond philosophy, logic is indispensable in everyday life, influencing how individuals process information, evaluate evidence, and make rational choices. Logical reasoning is crucial in decision-making, allowing people to weigh different options, consider consequences, and reach informed conclusions. Whether assessing financial investments, making healthcare choices, or engaging in political discussions, logical thinking helps individuals navigate complex issues without falling prey to misinformation or biases (Wong & Eurotech, 2021).

Logic also helps in providing clarity and coherence in discourse, thereby, enhancing communication. It is also used in debates, journalism and everyday conversations to improve persuasion and credibility. Moreover, one in recognising and avoiding fallacies which strengthens critical thinking and promotes constructive dialogue (Okolo & Nwaorgu, 2021).

Furthermore, logic is important in science and technology as scientific inquiry depends on logical reasoning to formulate hypotheses, conduct experiments and interpret data. Logical principles help to ensure that scientific conclusions are based on sound reasoning rather than fallacious assumptions (Zach, 2024).